DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1. TITLE MECHANICAL WORST CASE ANALYSIS REPORT OF DIMENSIONS AND TOLERANCES	**2. IDENTIFICATION NUMBER** DI-QCIC-80890

3. DESCRIPTION/PURPOSE

3.1 This report documents a mechanical worst case tolerance analysis performed on parts, subassemblies, assemblies and equipments.

4. APPROVAL DATE (YYMMDD) 890803	**5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)** G/Y254	**6a. DTIC APPLICABLE**	**6b. GIDEP APPLICABLE**

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

7.2 The documentation required by this DID is used to support the configuration end item product interchangeability requirements of DOD-D-1000B, Levels 2 and 3, DOD-STD-100C, MIL-STD-454J, Requirement 7, and DS-64 paragraph 5.

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8. APPROVAL LIMITATION	**9a. APPLICABLE FORMS**	**9b. AMSC NUMBER** G4800

10. PREPARATION INSTRUCTIONS

10.1 Format. The report may be in the contractor's format and shall be no larger than 8 1/2" by 11".

10.2 Content. The report shall contain the following:

10.2.1 Title page. The first page of the report shall be a title page and shall contain the following about the applicable part, subassembly, assembly or equipment analyzed:

a. Name.
b. Serial number.
c. Date of analysis.
d. Contract No.
e. Preparer of analysis.
f. Checker of analysis.

10.2.2 Table of contents. The second page of the report shall be a table of contents.

10.2.3 Body of report. The body of the report shall include the following:

10.2.3.1 Summary. A summary of the analysis, highlighting the potential problem areas.

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11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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OCK 7. APPLICATION/INTERRELATIONSHIP (continued)

.3 This DID is applicable whenever a contract requires mechanical designs with tolerance features.

7.4 This DID supersedes DI-E-5251A.

BLOCK 10. PREPARATION INSTRUCTIONS (continued)

10.2.3.2 Introduction. Introduction information shall include the following:

a. Title of each analysis performed.
b. Description of each analysis performed, or reference to the appropriate Government document.
c. Identification of the applicable parts, subassemblies, assemblies or equipments analyzed.
d. Requirements to be met.
e. Reason for performing analysis.

10.2.3.3 Data base. The data base shall include the following:

a. A list of all drawings/parts list.
b. Any referenced documents.
c. All form tolerances.
d. Justification for the use of any form tolerances.

10.2.3.4 Analysis. Analysis information shall contain the following:

a. Statement identifying the part, subassembly, assembly or equipment being analyzed.
b. Verification that a worst case tolerance analysis allows no excessive interferences, clearances, or excursions to exist that would cause failure.
c. Verification that the part, subassembly, assembly or equipment performs properly under all applicable environmental conditions, as specified on the DD FORM 1423.
d. A tolerance distribution for economical processing of the part, subassembly, assembly or equipment.
e. Applicable areas of engineering drawings or sketches to assist in the understanding of the analysis.
f. All calculations relevant to the analysis.
g. Any problems or potential problems with the part, subassembly, assembly or equipment.

10.2.3.5 Conclusion. The conclusion shall contain the following:

a. A summary of the analysis results.
b. Recommendations to correct any problems or potential problems noted during the analysis.

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